

REFRACTO

Enhancing Transparency in Loan Default Prediction Through Explainable AI

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INTRODUCTION

A tuned XGBoost ensemble can predict a default with strong accuracy. **On its own, it cannot explain why.**

Loan assessment was once a human-centric craft. As application volumes outgrew manual review, institutions turned to AI — and now face a paradox: transparent models like Logistic Regression are easy to justify but too simple for non-linear risk, while accurate ensemble models like XGBoost are “black boxes” no regulator can audit. Refracto resolves this by pairing a tuned XGBoost ensemble with SHAP post-hoc explanations, giving institutions both predictive power and a transparent reason for every decision.

OBJECTIVE

- Build a high-performance predictive model using ensemble learning for credit risk assessment.
- Implement a post-hoc explainability layer with SHAP to interpret the ensemble's black-box decisions.
- Evaluate the Hybrid XAI Framework's performance and transparency against traditional Logistic Regression.
- Design a web-based dashboard that visualizes risk scores and XAI justifications for credit evaluators.

PROBLEM STATEMENTS

- Low predictive sensitivity** — interpretable models like Logistic Regression miss non-linear risk, causing false rejections of creditworthy borrowers.
- Black-box complexity** — accurate ensemble models are mathematically opaque; “right to explanation” laws bar banks from deploying what they can't justify.
- Interpretability gap** — institutions are forced to choose between a transparent model that's too weak, or an accurate one that's inscrutable.
- A missing XAI layer** — no post-hoc tool turns a black-box “Default / No Default” flag into an auditable, per-feature diagnostic.

METHODOLOGY

MODEL & CALIBRATION

XGBoost (Tuned)	ROC-AUC 0.7194
Isotonic calibration	Brier 0.2145→0.1440
SHAP explanation	78 features, every case
Chosen threshold	0.4163 · LGD 63%

LGD (63%) is measured from 269K resolved charged-off loans; the threshold trades a little savings for 1.9× more recall than the pure cost-minimizer.

PIPELINE

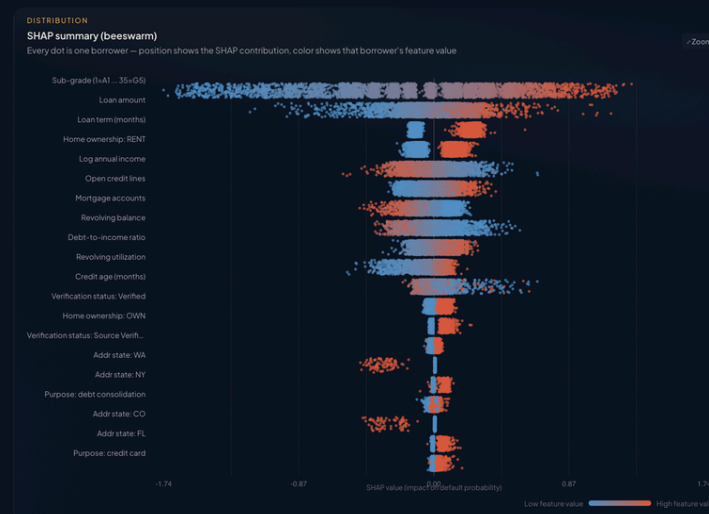
- LendingClub loans 2014–2020 (1.63M rows)
- EDA + feature engineering
- Stratified 80/20 split (seed 42)
- Isotonic calibration (out-of-fold)
- Train 8 models, baseline + tuned
- SHAP TreeExplainer + LinearExplainer
- LGD-weighted cost sweep → threshold
- FastAPI + React dashboard, live /predict

FEATURES

Refracto dashboard — live explainable credit scoring



Live Application Scoring — calibrated PD, model benchmark, expected loss



Explainability — how every feature swings risk across the whole test set

CONCLUSION

Refracto shows that predictive power and transparency are not mutually exclusive. The tuned **XGBoost ensemble** reaches a ROC-AUC of **0.7194**, **isotonic calibration** turns its raw score into a trustworthy probability, and **SHAP** explains every decision down to the feature — trading a small amount of savings at the chosen threshold to catch **1.9× as many defaults** as the pure cost-minimizer. Delivered as an interactive dashboard, it gives credit evaluators, risk managers and regulators a model they can both trust and explain — supporting SDG 8's goal of fair, inclusive access to financial services.